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Advances in Applied Clifford Algebras

Representations of Clifford Algebras on Function Spaces on the Cantor Set

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Abstract. We give a representation of the (infinite-dimensional) complex Clifford algebra on the Hilbert space of square-integrable complex-valued functions on the Cantor set.

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1. Introduction

We encountered the idea that the Clifford algebras could be represented on fractals in the paper [1], where the envisaged representation of Clifford algebras is undertaken via Cuntz algebras. In [2], representations of Cuntz algebras on fractal sets are investigated. In this note we want to give a direct realization of this pretty idea of representing Clifford algebras on fractals, without any use of Cuntz algebras.

We construct representations of the complex Clifford algebras $\mathbb{C}l_{2n}$ which naturally extend to a representation of the infinite dimensional complex Clifford algebra $\mathbb{C}l_\infty$. It has a more geometric flavour than the classical Fock representation. The method we use could be applied also to real Clifford algebras with mixed signature (see Remark 3.3).

We use the classical one-third Cantor set as a prototype of fractals. More generally, one could use also fractals $\mathcal{K}$, which can be realized as the attractor of a pair of contractions φ_0 and φ_1 on a metric space if the s-dimensional Hausdorff measure of $\varphi_0(\mathcal{K}) \cap \varphi_1(\mathcal{K})$ is zero, where s is the Hausdorff dimension of $\mathcal{K}$; for example $\mathcal{K} = [0, 1]$ with the two obvious contractions of scale 1/2 (for iterated functions systems see [3]).

In the following, $\mathcal{K}$ will denote the Cantor set, which we can think to be the attractor of the iterated functions system on $\mathbb{R}$, consisting of the functions $\varphi_0(x) = \frac{1}{3}x$ and $\varphi_1(x) = \frac{1}{3}x + \frac{2}{3}$: $\mathcal{K} = \varphi_0(\mathcal{K}) \cup \varphi_1(\mathcal{K})$ (with $\varphi_0(\mathcal{K}) \cap \varphi_1(\mathcal{K}) = \emptyset$).

$L^2\mathcal{K}$ will denote the complex Hilbert space of square integrable, complex-valued functions on $\mathcal{K}$ (with respect to the $\frac{ln2}{ln3}$-dimensional Hausdorff measure) and $B(L^2\mathcal{K})$ will denote the Banach algebra of bounded linear operators on $L^2\mathcal{K}$. A representation of a complex Clifford algebra $\mathbb{C}l_n$ on $L^2\mathcal{K}$ will be an algebra homomorphism $\mathbb{C}l_n \to B(L^2\mathcal{K})$.

2. Tilt- and Switch-Operators on $L^2\mathcal{K}$

We will define some operators on $L^2\mathcal{K}$ via symbolic dynamics on $\mathcal{K}$, which will be used to construct the representations of the complex Clifford algebras. We refer for symbolic dynamics to ([3],[4]) and recall a minimum necessary notation. Points of $\mathcal{K}$ have unique addresses which are infinite words $\omega_1\omega_2\cdots\omega_j\cdots$ with letters $\omega_j \in \{0,1\}$. If $x \in \mathcal{K}$ has the address $\omega_1\omega_2\cdots\omega_j\cdots$, then it holds

$$\{x\} = \bigcap_{j\geq 1} \varphi_{\omega_1} \circ \varphi_{\omega_2} \circ \cdots \circ \varphi_{\omega_j}(\mathcal{K}).$$

Identifying a point $x \in \mathcal{K}$ with its address, we will write $x = \omega_1\omega_2\cdots\omega_j\cdots$. For $j \in \mathbb{N}$, we define

$$\tilde{x}^j = \begin{cases} \omega_1\omega_2\cdots\omega_{j-1}0\omega_{j+1}\cdots & , \quad \text{for } \omega_j = 1 \\ \omega_1\omega_2\cdots\omega_{j-1}1\omega_{j+1}\cdots & , \quad \text{for } \omega_j = 0 \end{cases}$$

(flip-flap at j). We can decompose $\mathcal{K}$ with respect to the address-letter at a specific slot j :

$$\mathcal{K}_0^j := \{x \in \mathcal{K} \mid x = \omega_1\omega_2\cdots\omega_{j-1}0\omega_{j+1}\cdots\}$$

and

$$\mathcal{K}_1^j := \{x \in \mathcal{K} \mid x = \omega_1\omega_2\cdots\omega_{j-1}1\omega_{j+1}\cdots\}$$

with

$$\mathcal{K} = \mathcal{K}_0^j \cup \mathcal{K}_1^j.$$

Now we define two sets of operators T^j and S^j ($j \in \mathbb{N}$) on $L^2\mathcal{K}$. (As usual in functional analysis we operate with functions rather than equivalence class of functions on $\mathcal{K}$ for notational simplicity.) Let $f \in L^2\mathcal{K}$. Then we define

$$(T^j f)(x) = \begin{cases} f(x) & , \quad x \in \mathcal{K}_0^j \\ -f(x) & , \quad x \in \mathcal{K}_1^j \end{cases}$$

and

$$(S^j f)(x) = f(\tilde{x}^j) \text{ for } x \in \mathcal{K}.$$

(We call T^j the “tilt-”operators as they tilt the portion of graph on $\mathcal{K}_1^j$, and S^j the “switch-”operators as they switch the portions of graphs on $\mathcal{K}_0^j$ and $\mathcal{K}_1^j$.)

Example. We illustrate in the following figures a few stages of separation of $\mathcal{K}$ (Fig 1) and the effects of the operators T^1, S^1, T^2 and S^2 (Fig 3) on a sample function f with a graph as in Fig 2.

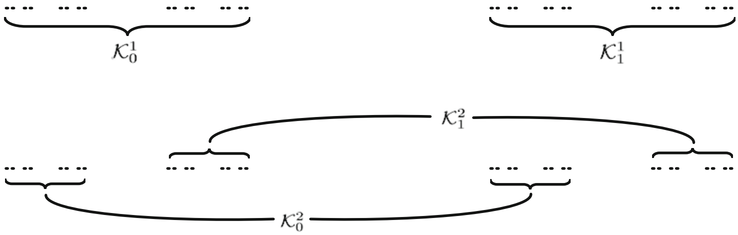

FIGURE 1. A few stages of separation of $\mathcal{K}$

FIGURE 2. The graph of f

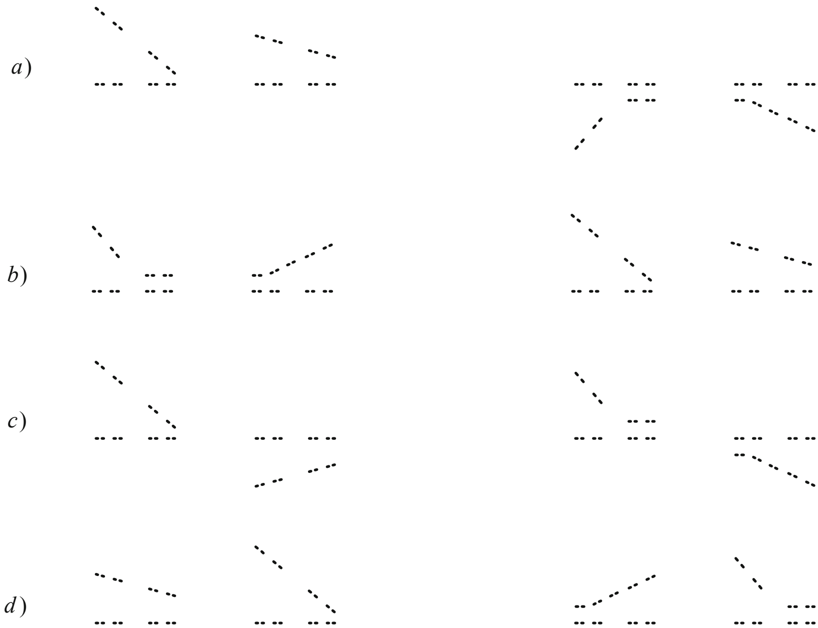

FIGURE 3. The graphs of (a) $T^1 f$, (b) $S^1 f$, (c) $T^2 f$, (d) $S^2 f$.

We now give a lemma about commutation properties of tilt- and switch-operators.

Lemma 2.1. *For $p, q \in \mathbb{N}$ the following equations hold:*

i) $T^p T^q = T^q T^p$

ii) $S^p S^q = S^q S^p$
iii) $T^p S^q = S^q T^p$ $(p \neq q)$
iv) $T^p S^p = -S^p T^p$

Proof. i) Let $f \in L^2\mathcal{K}$. Then,

$$(T^pT^q)(f)(x) = \begin{cases} (T^q f)(x) & , \quad x \in \mathcal{K}_0^p \\ -(T^q f)(x) & , \quad x \in \mathcal{K}_1^p \end{cases} \tag{2.1}$$

$$= \begin{cases} f(x) & , \quad x \in \mathcal{K}_0^p \cap \mathcal{K}_0^q \\ -f(x) & , \quad x \in \mathcal{K}_0^p \cap \mathcal{K}_1^q \\ -f(x) & , \quad x \in \mathcal{K}_1^p \cap \mathcal{K}_0^q \\ f(x) & , \quad x \in \mathcal{K}_1^p \cap \mathcal{K}_1^q \end{cases} \tag{2.2}$$

$(T^pT^q)(f)(x)$ has the same explicit expression.

ii) $(S^pS^q)(f)(x) = (S^q f)(\tilde{x}^p) = f(\widetilde{(\tilde{x}^p)}^q) = f(\widetilde{(\tilde{x}^q)}^p) = (S^p f)(\tilde{x}^q) = (S^qS^p)(f)(x)$.

iii)

$$(T^pS^q)(f)(x) = \begin{cases} (S^q f)(x) & , \quad x \in \mathcal{K}_0^p \\ -(S^q f)(x) & , \quad x \in \mathcal{K}_1^p \end{cases} \tag{2.3}$$

$$= \begin{cases} f(\tilde{x}^q) & , \quad x \in \mathcal{K}_0^p \\ -f(\tilde{x}^q) & , \quad x \in \mathcal{K}_1^p \end{cases} \tag{2.4}$$

For $p \neq q$ we have $x \in \mathcal{K}_0^p \Leftrightarrow \tilde{x}^q \in \mathcal{K}_0^p$ and $x \in \mathcal{K}_1^p \Leftrightarrow \tilde{x}^q \in \mathcal{K}_1^p$. Hence we can write,

$$(T^pS^q)(f)(x) = \begin{cases} f(\tilde{x}^q) & , \quad \tilde{x}^q \in \mathcal{K}_0^p \\ -f(\tilde{x}^q) & , \quad \tilde{x}^q \in \mathcal{K}_1^p \end{cases} \tag{2.5}$$

$$= (T^p f)(\tilde{x}^q) \tag{2.6}$$

$$= (S^qT^p)(f)(x). \tag{2.7}$$

iv)

$$(S^pT^p)(f)(x) = (T^p f)(\tilde{x}^p) \tag{2.8}$$

$$= \begin{cases} f(\tilde{x}^p) & , \quad \tilde{x}^p \in \mathcal{K}_0^p \\ -f(\tilde{x}^p) & , \quad \tilde{x}^p \in \mathcal{K}_1^p \end{cases} \tag{2.9}$$

$$= \begin{cases} f(\tilde{x}_p) & , \quad x \in \mathcal{K}_1^p \\ -f(\tilde{x}_p) & , \quad x \in \mathcal{K}_0^p \end{cases} \tag{2.10}$$

$$= \begin{cases} (S^p f)(x) & , \quad x \in \mathcal{K}_1^p \\ -(S^p f)(x) & , \quad x \in \mathcal{K}_0^p \end{cases} \tag{2.11}$$

$$= -(T^pS^p)(f)(x). \tag{2.12}$$

□

3. Representations of Complex Clifford Algebras on $L^2\mathcal{K}$

Let us denote the generators of the complex Clifford algebra $\mathbb{C}l_{2n}$ by e_j $(j = 1, 2, ..., 2n)$ (with ${e_j}^2 = 1$ and $e_j e_k = -e_k e_j$ for $j \neq k$). Let us map these generators in the following way into $B(L^2\mathcal{K})$:

$$\begin{array}{llll} \psi: & e_1 & \mapsto & T^1 \\ & e_2 & \mapsto & S^1 \\ & & \vdots & \\ (k>1) & e_{2k-1} & \mapsto & i^{(k+1)^2}(T^k T^{k-1} S^{k-1} T^{k-2} S^{k-2} \cdots T^1 S^1) \\ & e_{2k} & \mapsto & i^{(k+1)^2}(S^k T^{k-1} S^{k-1} T^{k-2} S^{k-2} \cdots T^1 S^1) \end{array}$$

(i is the complex i.)

Theorem 3.1. *ψ, induces an algebra homomorphism $\mathbb{C}l_{2n} \to B(L^2\mathcal{K})$, i.e. a representation of $\mathbb{C}l_{2n}$ on $L^2\mathcal{K}$.*

Proof. We have to see $(\psi(e_p))^2 = I$ and $\psi(e_p)\psi(e_q) = -\psi(e_q)\psi(e_p)$ for $1 \leq p \leq 2n,\ \ 1 \leq q \leq 2n,\ \ p \neq q$.

Let us first see that $(\psi(e_p))^2 = I$. Obviously, $(\psi(e_1))^2 = T^1T^1 = I$ and $(\psi(e_2))^2 = S^1S^1 = I$. Now let $p = 2k-1$ $(k > 1)$:

$$\begin{aligned} (\psi(e_{2k-1}))^2 &= (i^{(k+1)^2} T^k T^{k-1} S^{k-1} \cdots T^1 S^1)(i^{(k+1)^2} T^k T^{k-1} S^{k-1} \cdots T^1 S^1) \\ &= i^{2(k+1)^2}(-1)^{k-1}(T^k T^k T^{k-1} T^{k-1} S^{k-1} S^{k-1} \cdots T^1 T^1 S^1 S^1) \\ &\quad (\textit{by Lemma } 2.1) \\ &= (-1)^{(k+1)^2}(-1)^{k-1} I = (-1)^{k(k+3)} I = I. \end{aligned}$$

For $p = 2k$ we get

$$\begin{aligned} (\psi(e_{2k}))^2 &= (i^{(k+1)^2} S^k T^{k-1} S^{k-1} \cdots T^1 S^1)(i^{(k+1)^2} S^k T^{k-1} S^{k-1} \cdots T^1 S^1) \\ &= i^{2(k+1)^2}(-1)^{k-1}(S^k S^k T^{k-1} T^{k-1} S^{k-1} S^{k-1} \cdots T^1 T^1 S^1 S^1) \\ &= I. \end{aligned}$$

Let us now check the anti-commutativity relations. By Lemma 2.1, $\psi(e_1)\psi(e_2) = -\psi(e_2)\psi(e_1)$. Likewise, $\psi(e_1)$ and $\psi(e_2)$ anti-commute with all $\psi(e_j)$ for $j > 2$ by Lemma 2.1.

Now we consider various cases:

i) Let $p = 2k-1$, $q = 2l-1$ for $k > 1, l > 1$ and $k < l$.

$$\begin{aligned} \psi(e_p)\psi(e_q) &= (i^{(k+1)^2} T^k T^{k-1} S^{k-1} \cdots T^1 S^1)(i^{(l+1)^2} T^l T^{l-1} S^{l-1} \cdots T^1 S^1) \\ &= i^{(k+1)^2+(l+1)^2}(T^l T^{l-1} S^{l-1} \cdots T^{k+1} S^{k+1}) \\ &\quad (T^k T^{k-1} S^{k-1} \cdots T^1 S^1)(T^k S^k T^{k-1} S^{k-1} \cdots T^1 S^1) \\ &= i^{(k+1)^2+(l+1)^2}(-1)^{2k-1}(T^l T^{l-1} S^{l-1} \cdots T^1 S^1) \\ &\quad (T^k T^{k-1} S^{k-1} \cdots T^1 S^1) \\ &= -\psi(e_q)\psi(e_p). \end{aligned}$$

ii) Let $p=2k$, $q=2l$ for $k>1, l>1$ and $k<l$.

$$\begin{aligned}
\psi(e_p)\psi(e_q) &= (i^{(k+1)^2}S^kT^{k-1}S^{k-1}\cdots T^1S^1)(i^{(l+1)^2}S^lT^{l-1}S^{l-1}\cdots T^1S^1)\\
&= i^{(k+1)^2+(l+1)^2}(S^lT^{l-1}S^{l-1}\cdots T^{k+1}S^{k+1})\\
&\quad (S^kT^{k-1}S^{k-1}\cdots T^1S^1)(T^kS^kT^{k-1}S^{k-1}\cdots T^1S^1)\\
&= i^{(k+1)^2+(l+1)^2}(-1)^{2k-1}(S^lT^{l-1}S^{l-1}\cdots T^1S^1)\\
&\quad (T^kT^{k-1}S^{k-1}\cdots T^1S^1)\\
&= -\psi(e_q)\psi(e_p).
\end{aligned}$$

iii) Let $p=2k-1<q=2l$, $k>1, l>1$.

$$\begin{aligned}
\psi(e_p)\psi(e_q) &= (i^{(k+1)^2}T^kT^{k-1}S^{k-1}\cdots T^1S^1)(i^{(l+1)^2}S^lT^{l-1}S^{l-1}\cdots T^1S^1)\\
&= i^{(k+1)^2+(l+1)^2}(S^lT^{l-1}S^{l-1}\cdots T^{k+1}S^{k+1})\\
&\quad (T^kT^{k-1}S^{k-1}\cdots T^1S^1)(T^kS^k\cdots T^1S^1)\\
&= i^{(k+1)^2+(l+1)^2}(-1)^{2k-1}(S^lT^{l-1}S^{l-1}\cdots T^1S^1)\\
&\quad (T^kT^{k-1}S^{k-1}\cdots T^1S^1)\\
&= -\psi(e_q)\psi(e_p).
\end{aligned}$$

The remaining cases are seen similarly. $\square$

Remark 3.2. Let us denote the above representation $\psi : \mathbb{C}l_{2n} \to B(L^2\mathcal{K})$ by ψ_{2n} and consider the commutative diagram

$$\begin{tikzcd}
\mathbb{C}l_2 \arrow[r] \arrow[drr, "\psi_2"'] & \mathbb{C}l_4 \arrow[r] \arrow[dr, "\psi_4"] & \cdots \arrow[r] & \mathbb{C}l_{2n} \arrow[r] \arrow[dl, "\psi_{2n}"] & \cdots, \\
& & B(L^2\mathcal{K}) & &
\end{tikzcd}$$

where the horizontal mappings are the natural embeddings. As the diagram is commutative, it yields a representation of $\mathbb{C}l_\infty$ on $L^2\mathcal{K}$.

Remark 3.3. The real Clifford algebra $\mathbb{C}l_{p,q}$ (with $e_i^2=1$ for $i=1,2,...,p$ and $e_i^2=-1$ for $i=p+1,p+2,...,p+q$) can be represented on the Hilbert space of square integrable real-valued functions on the Cantor set by the following map on generators:

$$\begin{array}{lcl}
e_1 & \mapsto & S^1\\
e_2 & \mapsto & S^2T^1\\
e_3 & \mapsto & S^3T^2T^1\\
 & \vdots & \\
e_p & \mapsto & S^pT^{p-1}T^{p-2}\cdots T^2T^1\\
e_{p+1} & \mapsto & S^1T^1\\
e_{p+2} & \mapsto & S^2T^2T^1\\
e_{p+3} & \mapsto & S^3T^3T^2T^1\\
 & \vdots & \\
e_{p+q} & \mapsto & S^qT^qT^{q-1}T^{q-2}\cdots T^2T^1
\end{array}$$

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